NISHANT LAKHARA

Staff Engineer/ Principal Engineer

+919028590871 • nishant .lakhara.280389@gmail.com • Bangalore, Karnataka

Summary

An experienced Principal / Lead Engineer with over 14+ years of experience in developing scalable, high-performance solutions across various domains, including eCommerce and finance. I have worked with clients such as Apple, Morgan Stanley, Nomura and Walmart. My expertise lies in Core Java, Spring Boot, Microservices, and cloud technologies like AWS. I have a proven track record of optimizing software, leading development teams, and delivering high-quality solutions. I am committed to continuous learning and improving development processes.

Skills

Core Java (Expert), Spring Boot (Intermediate), Microservices, Python, Architecture, Design, Data Structures, Optimization, RabbitMQ, Apache Kafka, Hazelcast, Hazelcast Compute, GemFire, SQL, MongoDB, HBase, CI/CD, Jenkins, Nginx, Docker, Ansible, Unix, Shell Scripting, Terraform, Sumo Logic, React JS (Beginner), Vue JS (Beginner), Leadership, Mentoring

AI Adoption (Intermediate): AI/ML, Generative AI, Diffusion Models, Github copilot, chatgpt, opus etc

Key Achievements (Technical)

Kafka Integration

Integrated and deployed a secure **Apache Kafka (MSK)** architecture with **SSL** and **SASL_SSL** security mechanisms, ensuring encrypted communication and secure client connections. Integrated **Kafka** for real-time data streaming, significantly enhancing system scalability and enabling efficient, high-throughput data processing across distributed services. Added **Kafka Debezium Connector** integration with AWS RDS for real time client notifications to any update in the entities.

Data Synchronization Optimization

Optimized Data Synchronization: Integrated NO SQL **HBase (MAPR)** built on top of HDFS (Hadoop) as a third-layer caching solution in a high-performance orchestration system, reducing data synchronization time from over 2 hours to near-instantaneous, enhancing system efficiency and scalability by eliminating the secondary SQL cache layer. Added custom indexes in the application for faster retrieval of data and reduced hot spotting by careful key analysis in a clustered environment with nodes and replicas of HBASE.

Automation & Performance Optimization

Automated critical processes including commit, build, test, and deployment using Go pipelines and Ansible scripts, improving development efficiency and reducing deployment issues. Simultaneously, led JVM tuning efforts by transitioning from Mark Sweep to G1 GC, significantly reducing GC pauses and optimizing memory management. Also fine-tuned Hazelcast cache clusters and optimized data flow, resulting in enhanced system performance, scalability, and reliability under high loads.

AWS Cost Optimization

Reduced **AWS infrastructure costs by 20-30%** through strategic optimization of services like RDS, EC2, and Kafka, and fine-tuning CI/CD pipelines.

Centralized Kafka Connect

Architected and centralized Kafka Connect–based integrations replacing scheduler-based pipelines, enabling scalable event-driven data movement.

Platform Modernization / Migration

Led modernization of ~25 microservices at Walmart by upgrading from Spring Boot 2.7 to 3.x and migrating search infrastructure from Elasticsearch to OpenSearch using a centralized dependency governance framework. This improved security posture, reduced dependency conflicts, strengthened long-term platform maintainability, and introduced engineering guardrails such as contract testing and rate limiting to enhance deployment safety and reliability.

AI-Augmented Engineering Adoption

Leveraged AI-assisted development tools including GitHub Copilot, Windsurf Editor, and advanced LLMs such as Claude Opus to accelerate development, code reviews, and documentation generation across microservices. **Currently undergoing advanced AI/ML training from Indian Institute of Technology Kharagpur**, applying learned concepts to improve developer productivity and engineering workflows.

Experience

Walmart Global Tech India

Staff Engineer

Bangalore 02/2025 - Present

- Leading backend architecture initiatives for real estate and supply chain platforms.
- Modernized 25+ Spring Boot services from 2.7 → 3.x, improving dependency governance and security.
- Designed centralized Kafka Connect integrations, replacing legacy batch pipelines.
- Implemented platform guardrails including contract testing, rate limiting, and startup probes improving production stability.
- Delivered key supply chain systems including warehouse replenishment, order tracking, and order recovery frameworks.
- Reduced platform infrastructure costs by ~25% through cloud optimization.
- Mentored a team of 12 developers including SSEs and below.

Experience

Insightsoftware
Hyderabad

Principal Engineer 09/2023 - 01/2025

- Managed a team of 5 developers to build FastPost, a no-code, rule-driven accounting software utilizing Java Spring Boot, Vue.js, and AWS, while also integrating Hazelcast Cache and Jet Pipelines for scalable real-time data processing.
- Developed and ran end-to-end performance testing environments using Docker, Terraform, and Ansible, provisioning EC2 instances, RDS, and MSK Kafka Clusters for client deployments while optimizing AWS infrastructure, reducing AWS costs by 20-30%.
- Developed and deployed secure Apache Kafka (MSK) architecture with SSL and SASL_SSL security mechanisms for encrypted communication.
- Collaborated with Business Analysts, Product Owners, Leads to implement client requirements in banking products.

Oportun (Intelizest Consultency)
Remote

Senior Backend Engineer 03/2022 - 07/2023

- Ledend-to-end feature development for a loan application microservices, including requirement gathering, coding, debugging,and deployment in the CI/CD pipeline.
- Collaborated with Product owners, architects, peers and leads on enhancement developments and bug fixes using TAR documents.
- Worked on Docmosis (DocuSign) and generalized 124 templates into just 2, increasing productivity.
- Conducted code/PR reviews, mentored a team of 2-3 engineers, and provided KT sessions for new joiners.

Morgan Stanley (Wissen Technology)
Mumbai

Senior Consultant 01/2020 - 03/2022

Company Description

- Migrated legacy monolithic views to microservices architecture, integrating asynchronous messaging with TIBCO.
- Automated a 4-5 hour manual process using a Java Spring Boot framework, reducing verification time to minutes.
- Developed an exception tracking framework for Equity risk reporting using Python.
- Generated around 100 reconciliation reports for downstream flows using in-house tools.

Nomura (AgiliteGroup)
Mumbai

Senior Application Developer 02/2017 - 01/2020

- Createdand optimizedWeb/REST services for risk calculations, involving technical design, development, deployment,testing, performance tuning, scripting, and automation.
- Designed and optimized orchestration architecture with Spring Boot and asynchronous messaging, managing data flow from RDS to Risk Engine and Aggregator. Exposed client APIs to ensure transactional consistency across services, following a SAGA-like pattern.
- Optimized data synchronization by integrating HBase on MapR as a second layer (first being Gemfire cache), replacing SQL. Reduced sync time from over 2 hours to near-instantaneous after processing millions of records. Disabled SQL layer post-production testing to further improve performance.
- Established a Prod Parallel environment using Autosys and Shell scripting, optimizing testing before moving code to production.

Apple (Appperfect Corporation, Wipro)
Udaipur

Senior Software Engineer 02/2014 - 01/2017

- Workedon Javaweb services (Order Service, Payment Service) for the Apple Online Store and mentored a team of5 engineers through development, deployment, and troubleshooting.
- Automated commit, build, test, and integration processes using Go pipelines, streamlining deployments and ensuring high code quality.
- Automated RabbitMQ deployment to Linux machines with Ansible, improving update and verification efficiency.
- Developed clustering scripts for Hazelcast, RabbitMQ, and Solace, enhancing system scalability and reliability.
- Built and tuned a Hazelcast cluster (3 nodes), optimizing GC properties and transitioning to G1 GC, reducing GC pauses and improving memory management for large heaps.
- Led a POC on Apache Traffic Server and NGINX reverse proxies, evaluating load balancing and traffic management solutions.
- Led a team of 5 in various development tasks and guided them through the deployment process.

Geometric Limited
Pune

Software Engineer 08/2011 - 01/2014

- Gathered requirements and prepared Technical Approach documents for system integration.
- Enhanced system efficiency by {15%} through development of {5} integration connectors using Core Java.

Education

Sir Padampat Singhania University — Udaipur

BTech inComputerScience — 8.8 CGPA — 08/2007 - 06/2011

Gold Medalist: Batch Topper

Alok Senior Secondary School — Udaipur, Rajasthan

CBSE 12th Board in Science and Maths — 86.4% — 01/2006 - 01/2007

Training / Courses

Oracle Certified Java Professional

Gen AI Training at Walmart by IIT Kharagpur